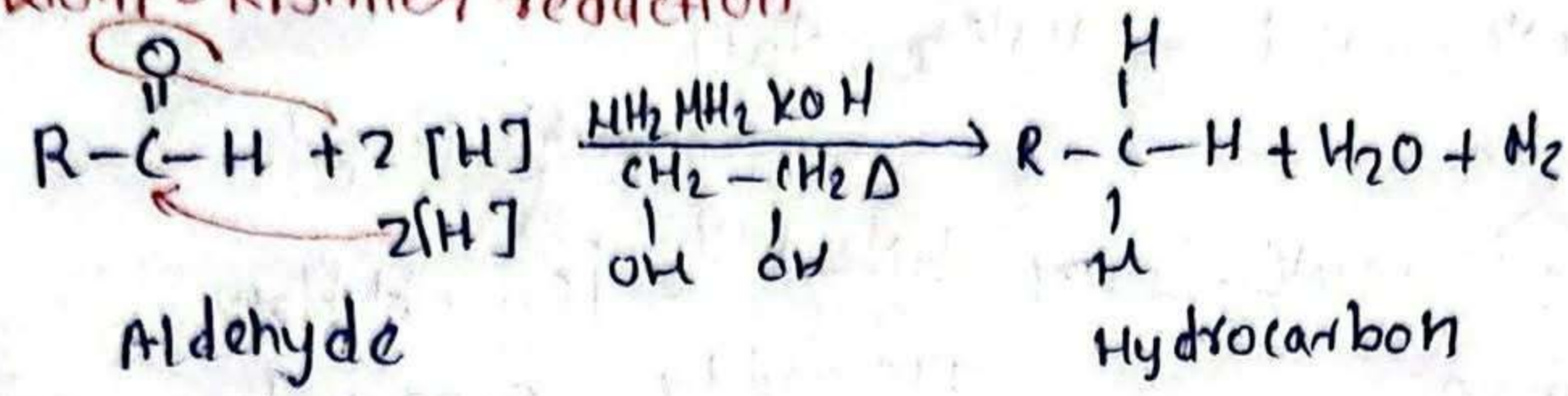
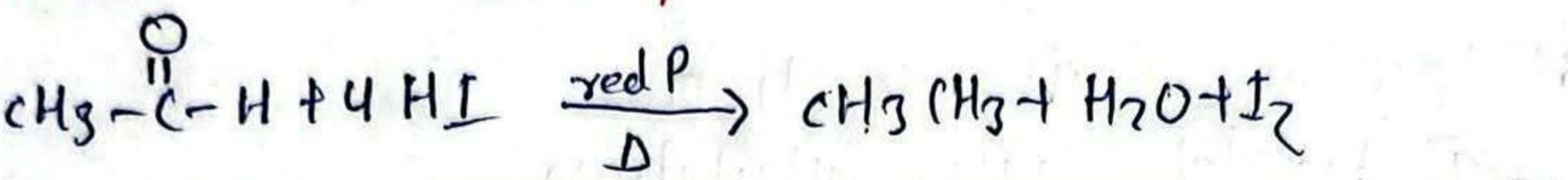
42. Wolff - kishner reduction

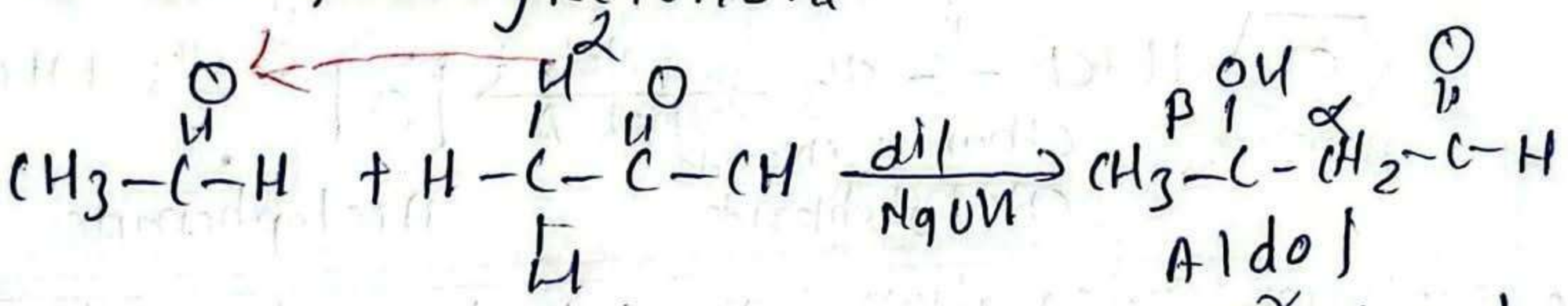


NOTE Reduction by HI & red p

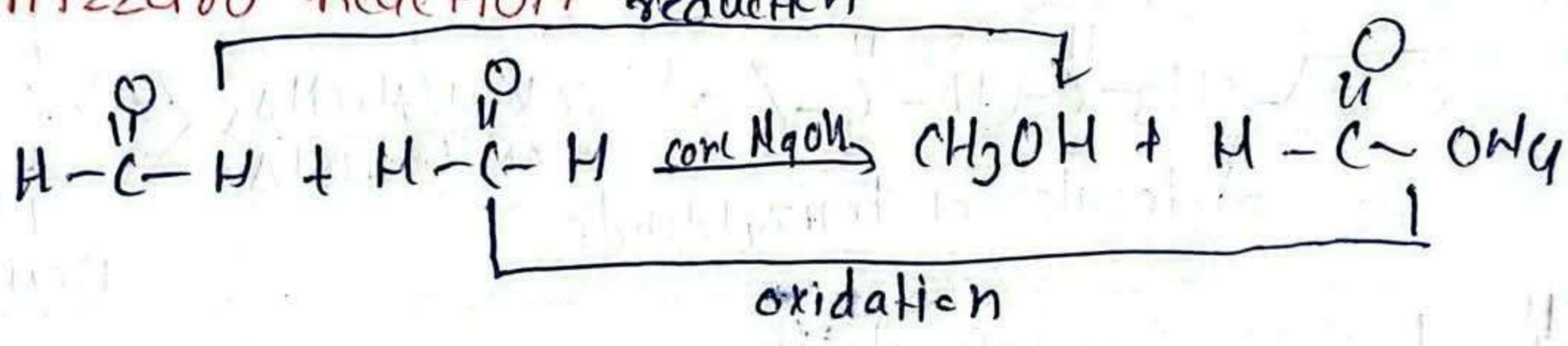


43 Aldol condensation

In this reaction two molecules of ALDEHYDE or ketone condense in presence of dil alkali (Dil NaOH, Na2CO3 etc) to form hydroxyladehyde or De-hydroxyketonone.



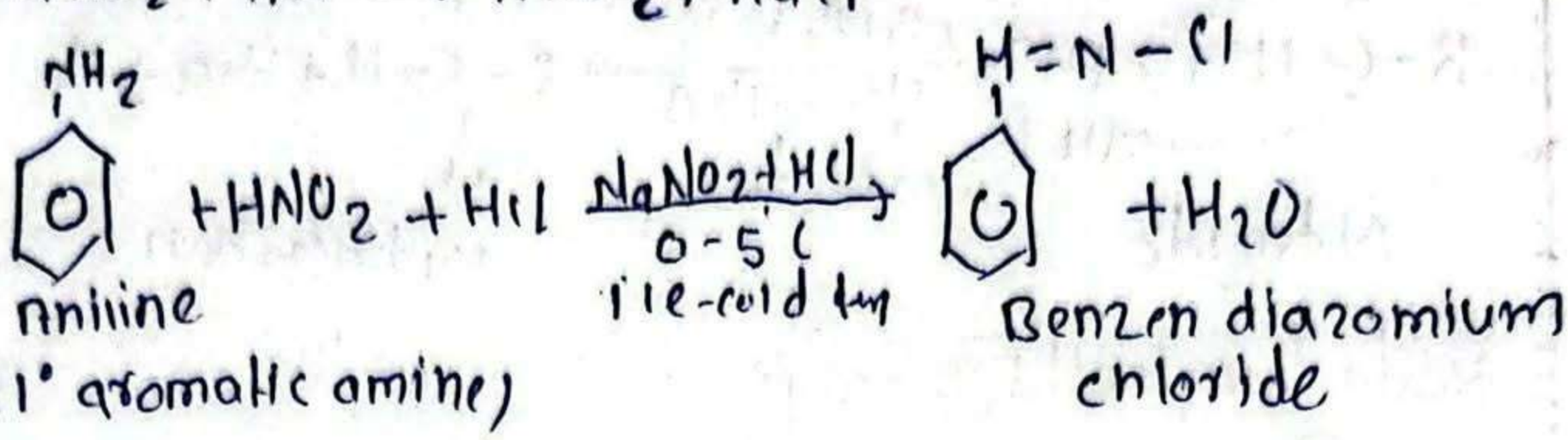
44 Cannizzaro Reaction



↳ Aldehydes without α-hydrogen when treated with concentration NaOH / KOH undergo Self oxidation and reduction. one molecules oxided to carboxylic acid and the other is reduced to alcohol.

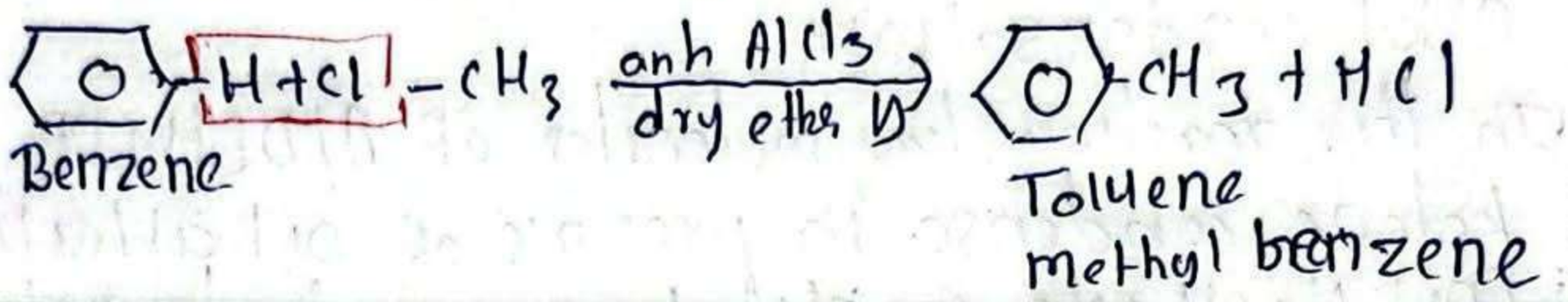
38 Diazontization reaction

$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$

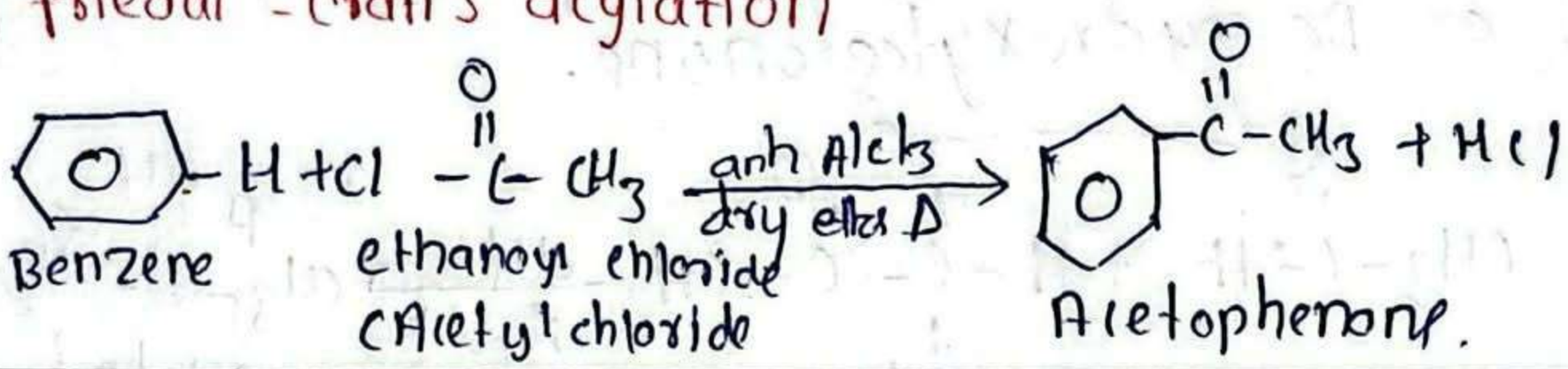


39 Friedal - Craff's reaction

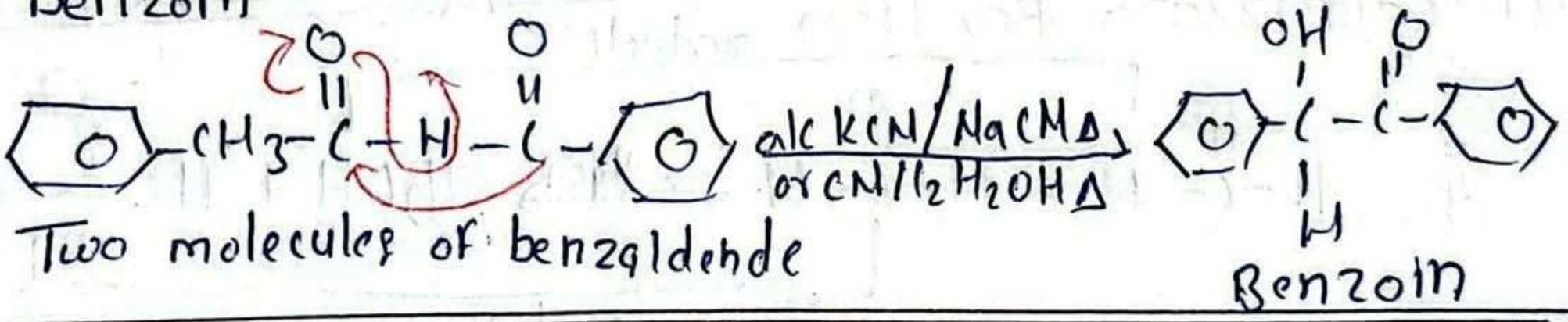
a] friedal - Craff's alkylation



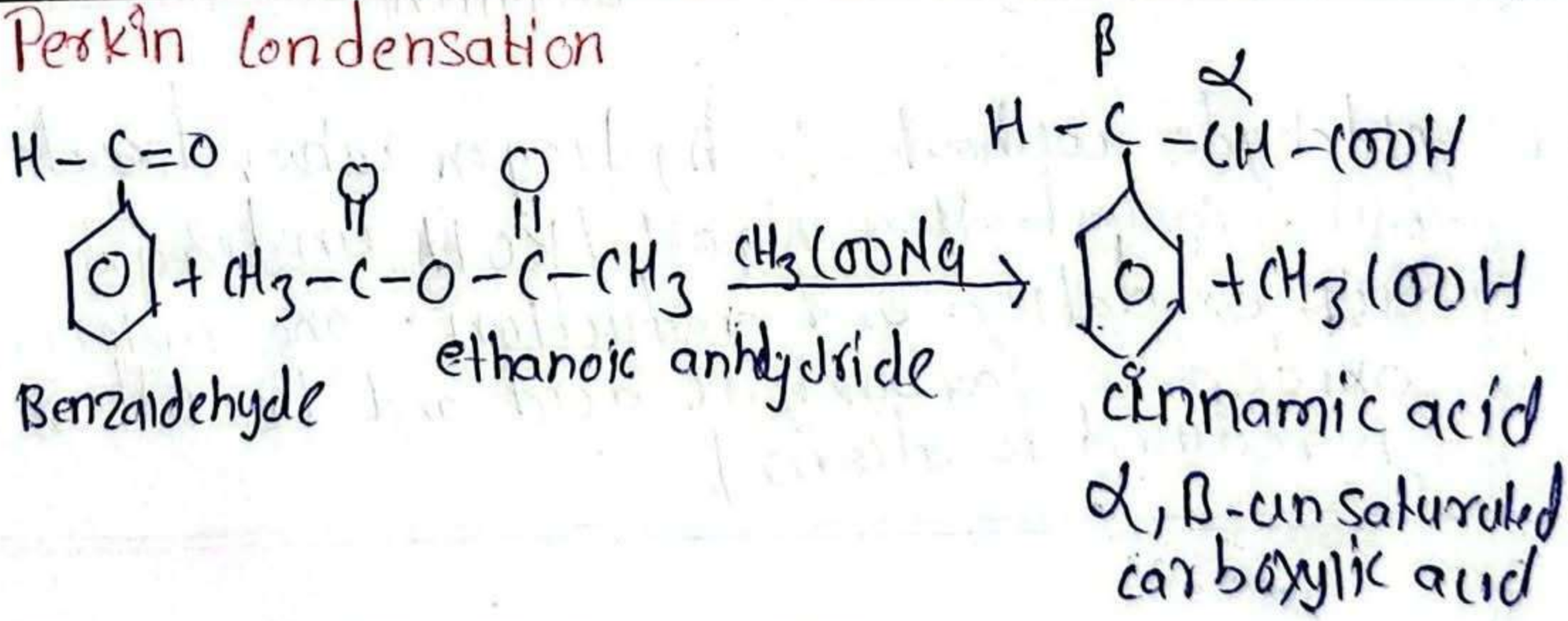
b] friedal - Craff's acylation



40 Benzoin condensation ÷ Two molecules of benzaldehyde condense in the presence of a/c KCN or NaCN form benzoin



41 Perkin condensation
2082



33 clemmensen's reducion:

$$CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + 2[H] \xrightarrow[\text{con HCl}]{Zn-Hg} CH_3-\underset{\underset{H}{|}}{\overset{H}{\overset{|}{C}}}-CH_3 + H_2O$$

propanone 2[H] propane

---

34 Hell-volhard-zelinsky (HVZ) reaction.

$$CH_3COOH \xrightarrow[-HCl]{Cl_2, P} ClCH_2COOH \xrightarrow[-HCl]{Cl_2 P} CH_2ClCOOH$$

Acetic acid — chloroacetic acid — Dichloroacetic acid

$$\xrightarrow[-HCl]{Cl_2 P} Cl_2CHCOOH$$

Trichloroacetic acid.

---

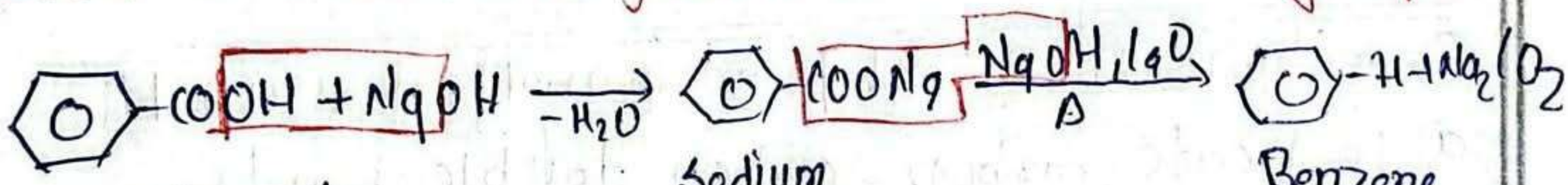
35 claisen condensation : Ester having α-hydrogen undergoes self condensation in the present of strong base like sodium ethoxide to give β-keto ester.

$$CH_3-\overset{O}{\overset{\|}{C}}-OC_2H_5 + H-CH_2-COOC_2H_5 \xrightarrow[-C_2H_5OH]{+C_2H_5ONa} CH_3-\overset{O}{\overset{\|}{C}}-CH_2COOC_2H_5$$

ethyl acetate — ethyl aceto-acetate (β-keto ester)

---

36 Soda-lime decarboxylation (loss of -COOH group)



Benzoic acid — Sodium benzate — Benzene

---

37 Baeyer's reagent test (test for unsaturation) : alkaline $KMnO_4$ ($KMnO_4 + KOH$)

$$KMnO_4 + KOH \longrightarrow K_2MnO_4 + H_2O + [O]$$

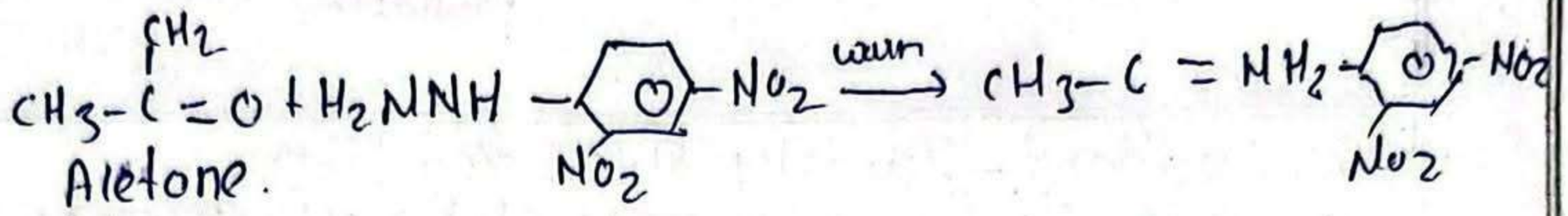
pink — colourless

$$CH_2=CH_2 + H_2O + [O] \longrightarrow CH_2-CH_2$$

ethene (unsaturated) — (2OH) — OH OH

29) 2,4 - DNP test or 2,4 - DNPH test (NEB 2082)
2082

to identify carbonyl compounds (aldehyde & ketone) or to distinguish carbonyl compounds from non-carbonyl compounds.



30 Tollen's (reagent) test or silver mirror test.

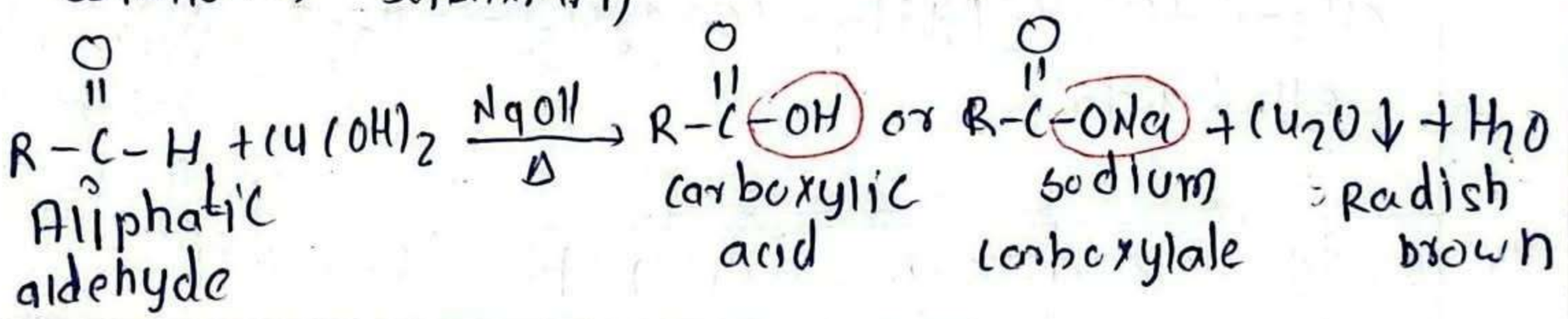
$AgNO_3 + NH_4OH \longrightarrow AgOH \downarrow + NH_4NO_3$

$AgOH + 2NH_4OH \longrightarrow [Ag(NH_3)_2]OH + 2H_2O$

31 Fehling's solution test : to distinguish aliphatic aldehydes from ketones or aromatic aldehyde

$CuSO_4 + 2NaOH \longrightarrow Cu(OH)_2 \downarrow + Na_2SO_4$

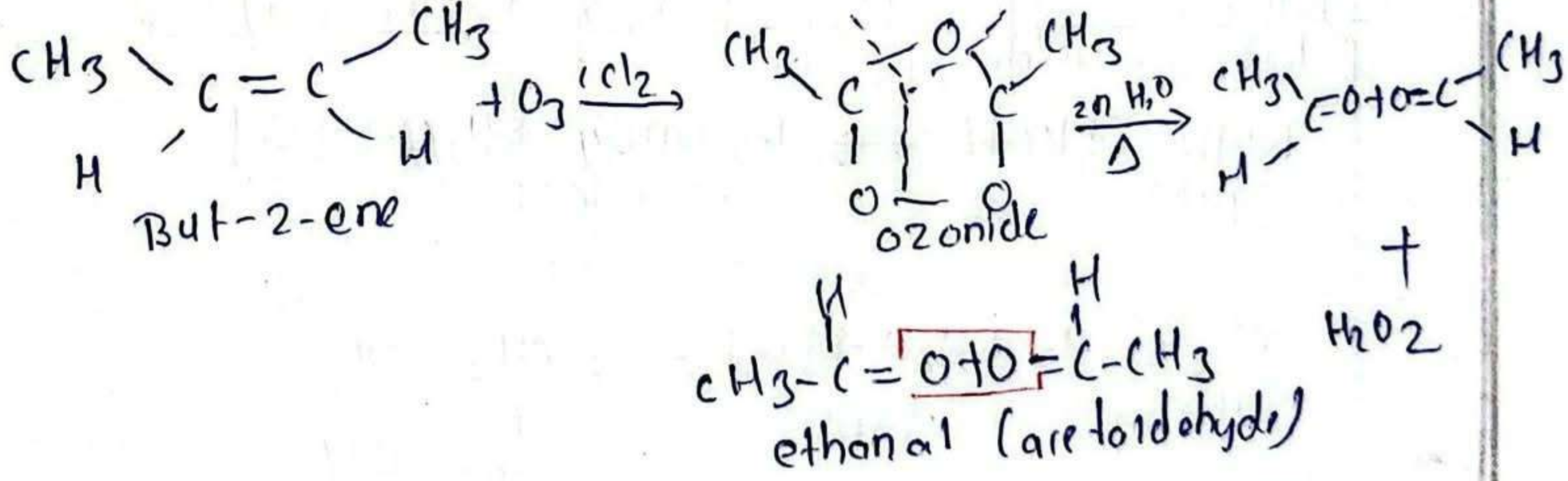
Fehling's solution (I) — Fehling solution (II) — blueish white.



32 Ozonolysis [NEB 2081] and ozanolysis is used ---?

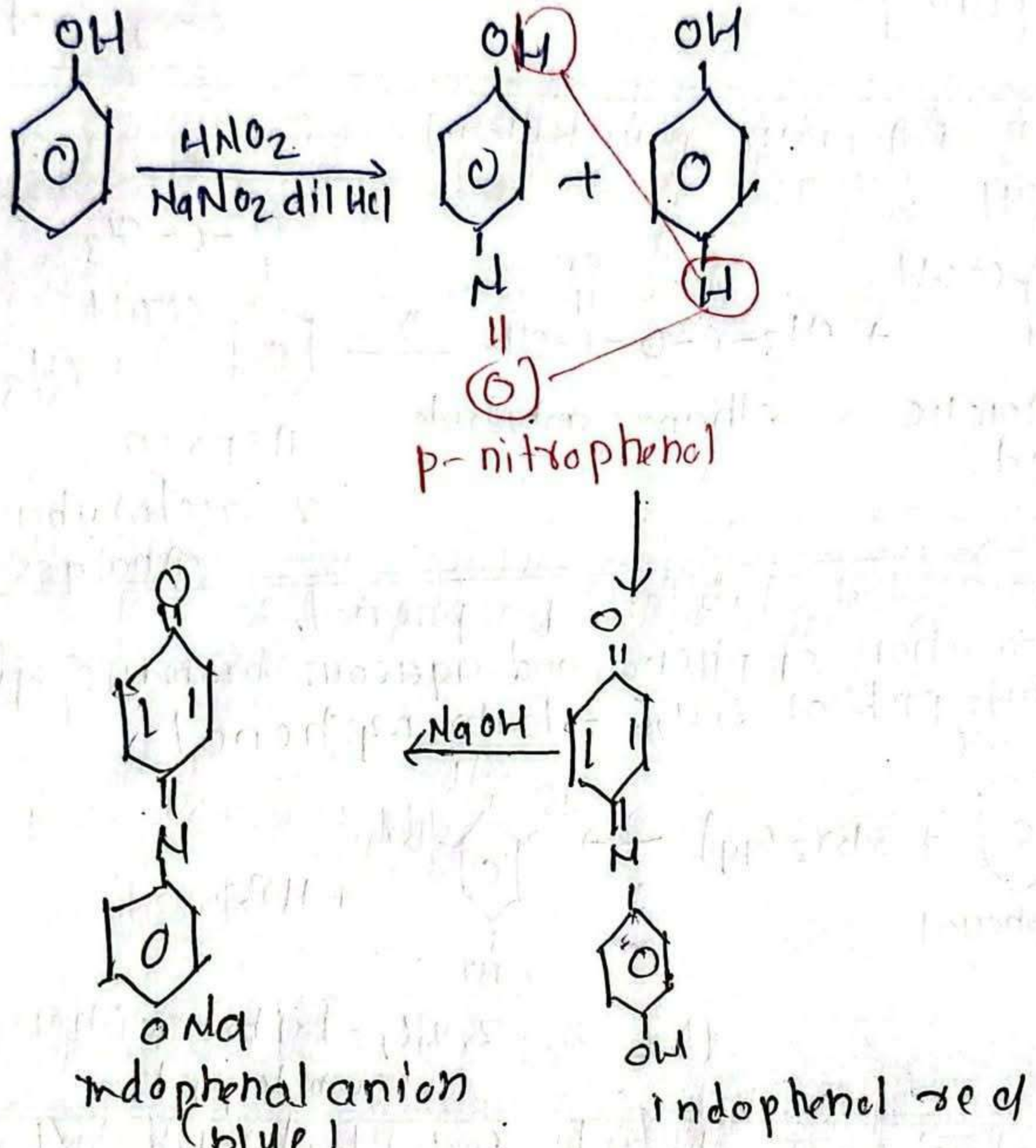
a] to locate carbon - carbon double bonds

b] to prepare aldehydes & ketones.

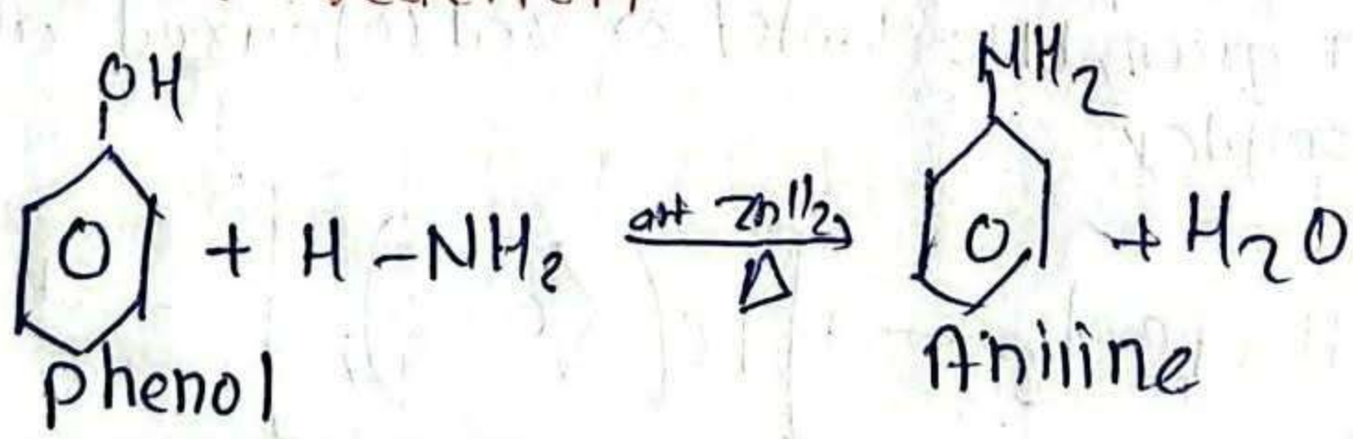


26] libermann's test (lab test for phenol)

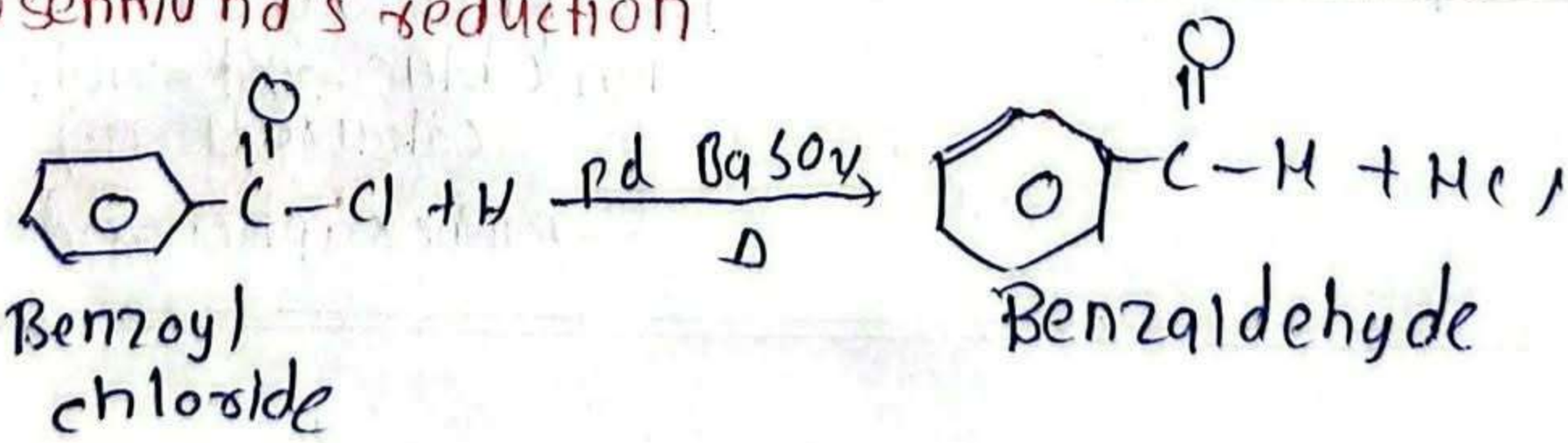
→ phenol reacts with nitrous acid ($NaNO_2 + H_2SO_4$) in the presence of conc H_2SO_4 is given brown and red colour on dilution which become blue or green on adding NaOH solution.



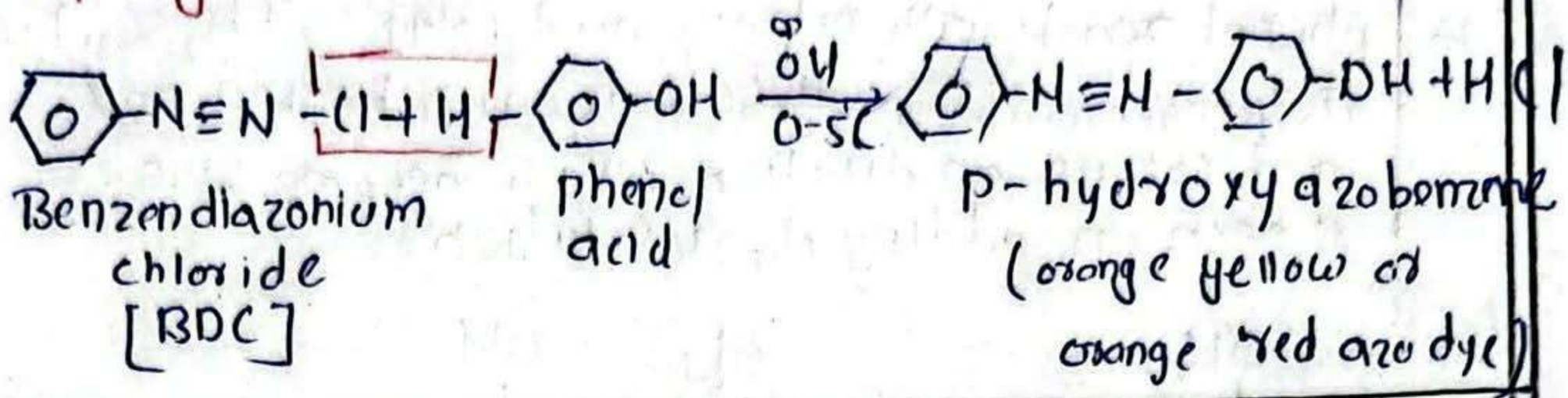
27] Bucherer reaction



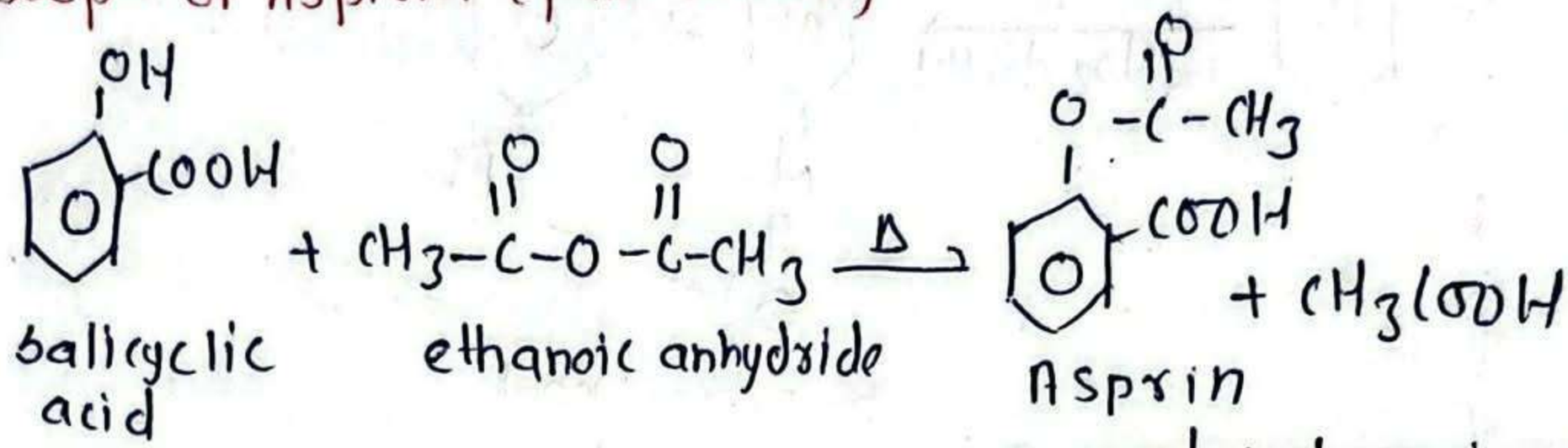
28] Rosenmund's reduction



23 Coupling reaction:

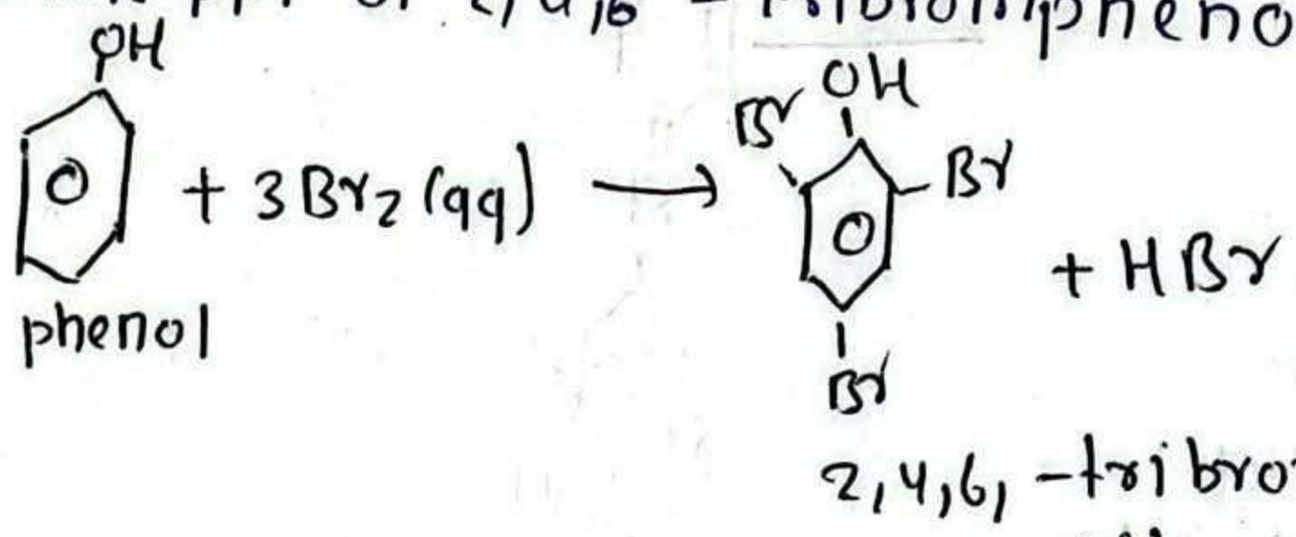


prepn of Aspirin (pain killer)

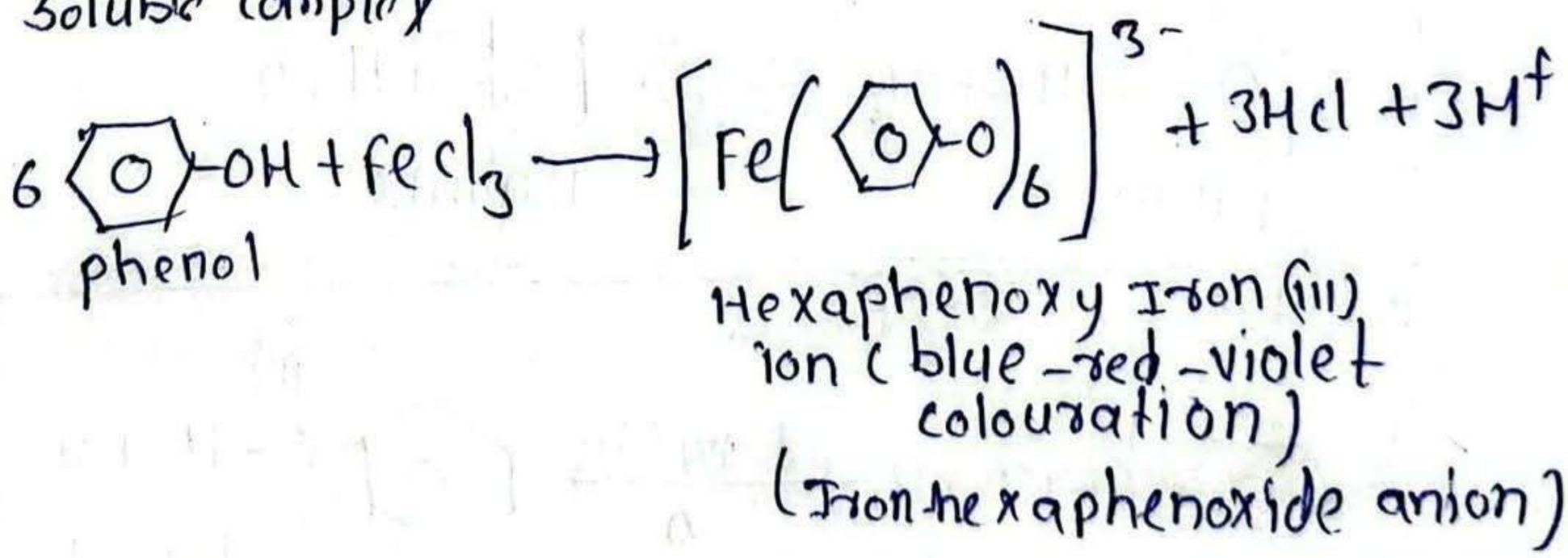


24 Bromine test (lab test) for phenol

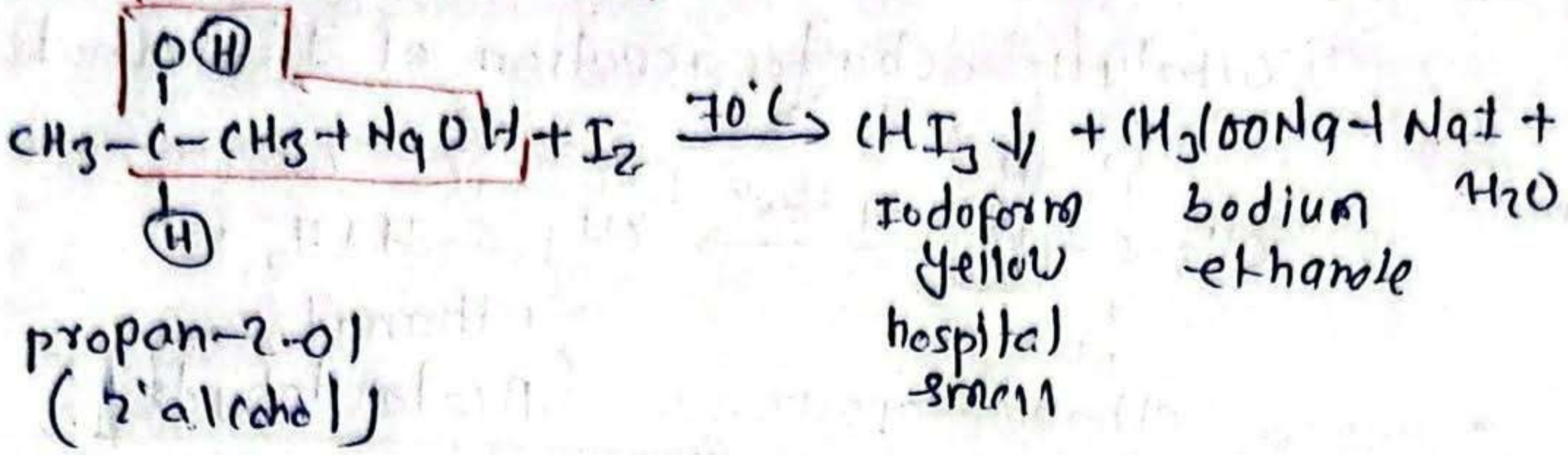
The reaction of phenol and aqueous bromine gives a white ppt of 2,4,6 - tribromphenol



25 Neutral ferric chloride ($FeCl_3$) test ÷ lab test for phenol (enolic group) ÷ phenol reacts with $FeCl_3$ solution to gives ~~green~~ green, blue-violet or red coloured water soluble complex



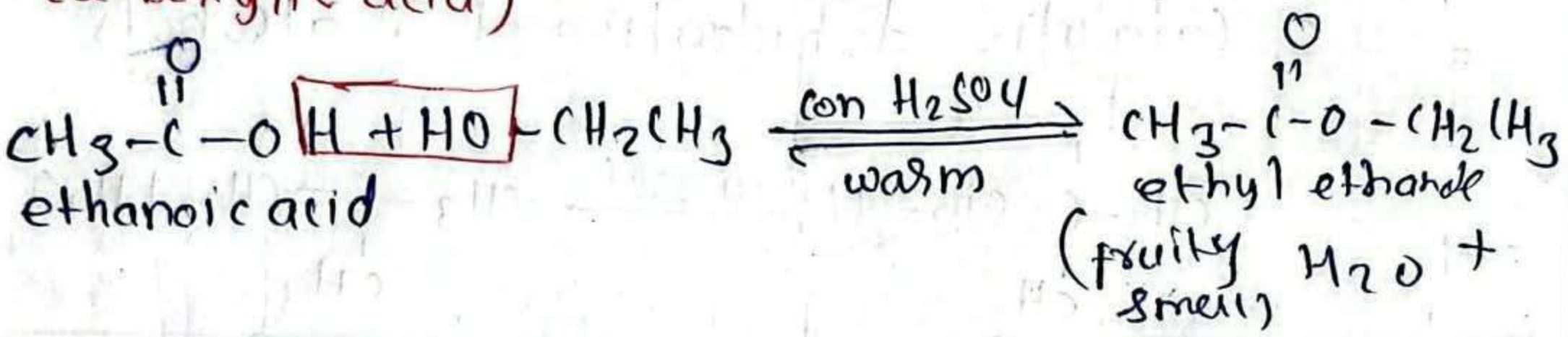
Iodoform test for propan-2-ol



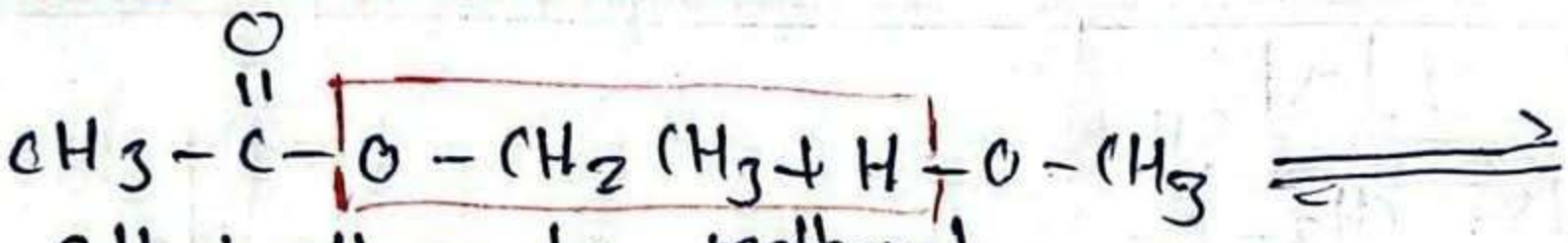
Iodform say R कुना नाइ water

$CHI_3 + RCOONa + NaI + H_2O$

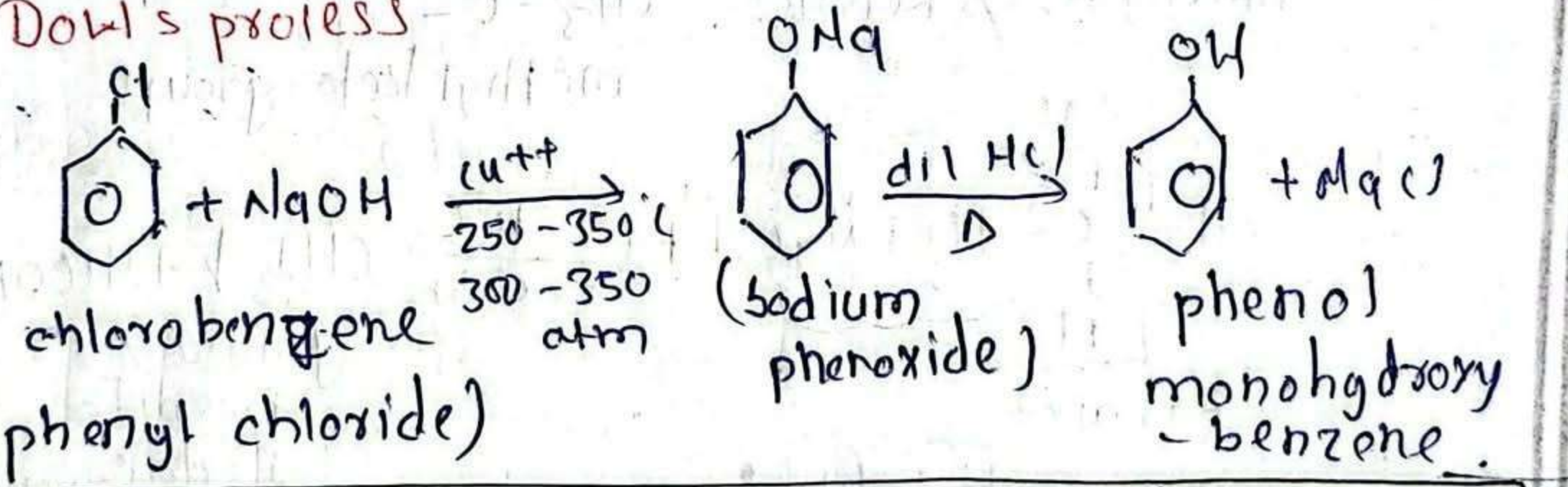
Esterification reaction (lab test for alcohol or carboxylic acid)



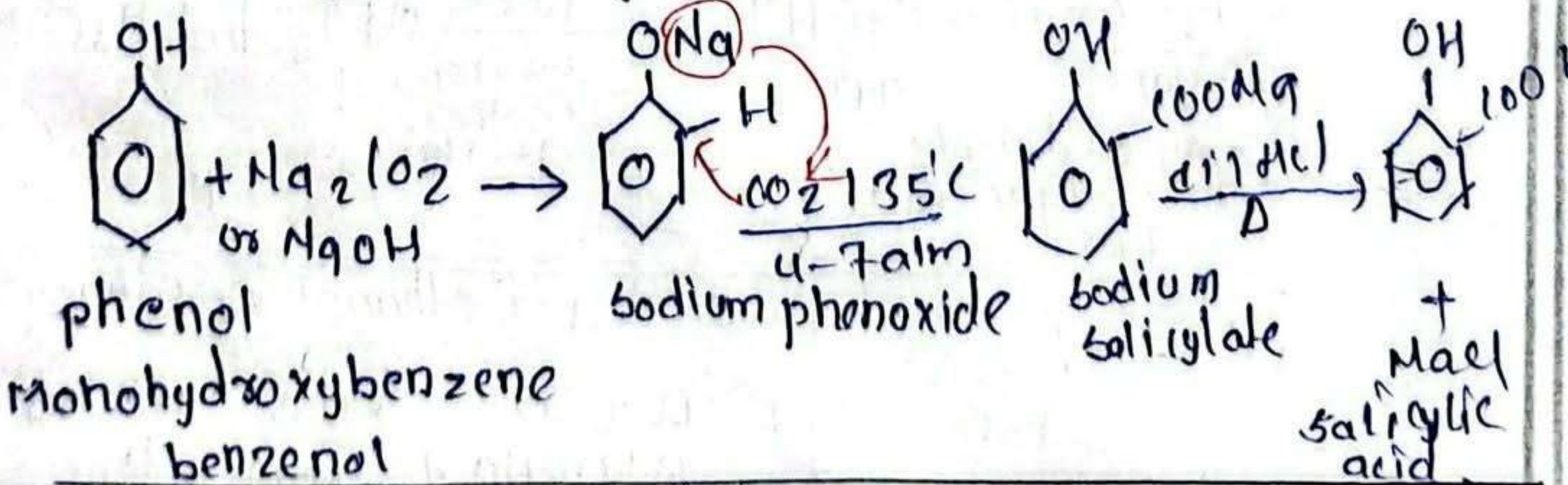
Trans-esterifaction



Dow's process

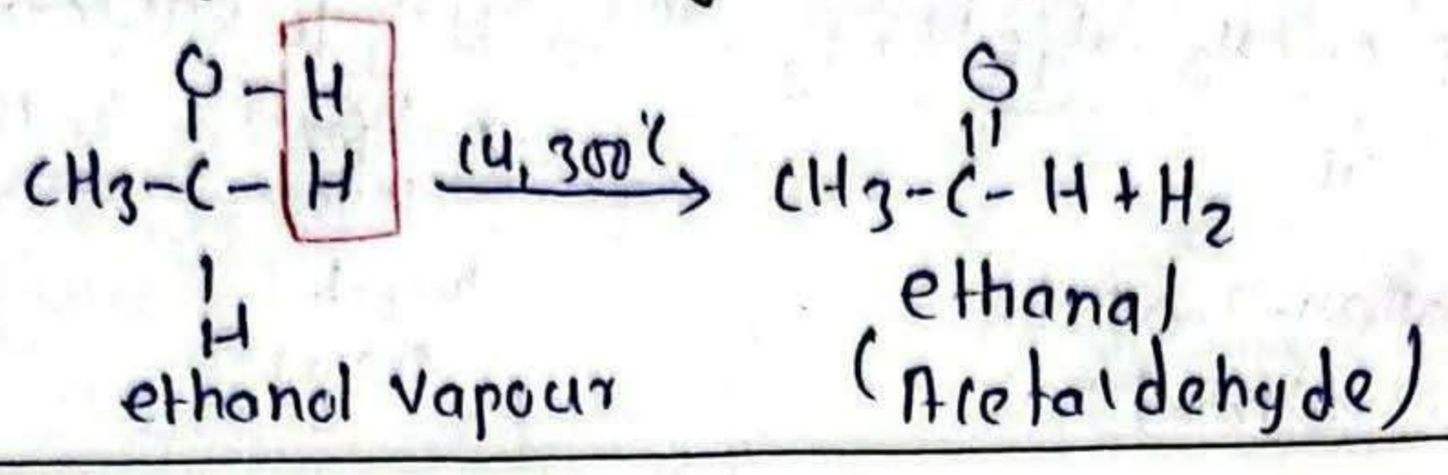


Kolbe's carboxylation reaction (Introduction of carboxylic acid group)

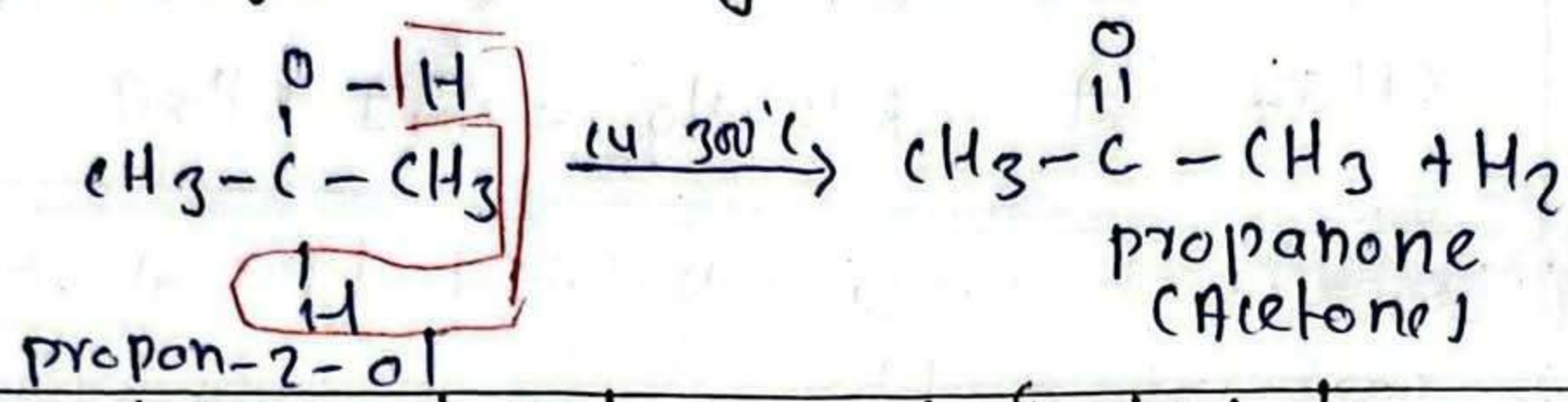


17] Catalytic dehydrogenation of 1°, 2° & 3° alcohols

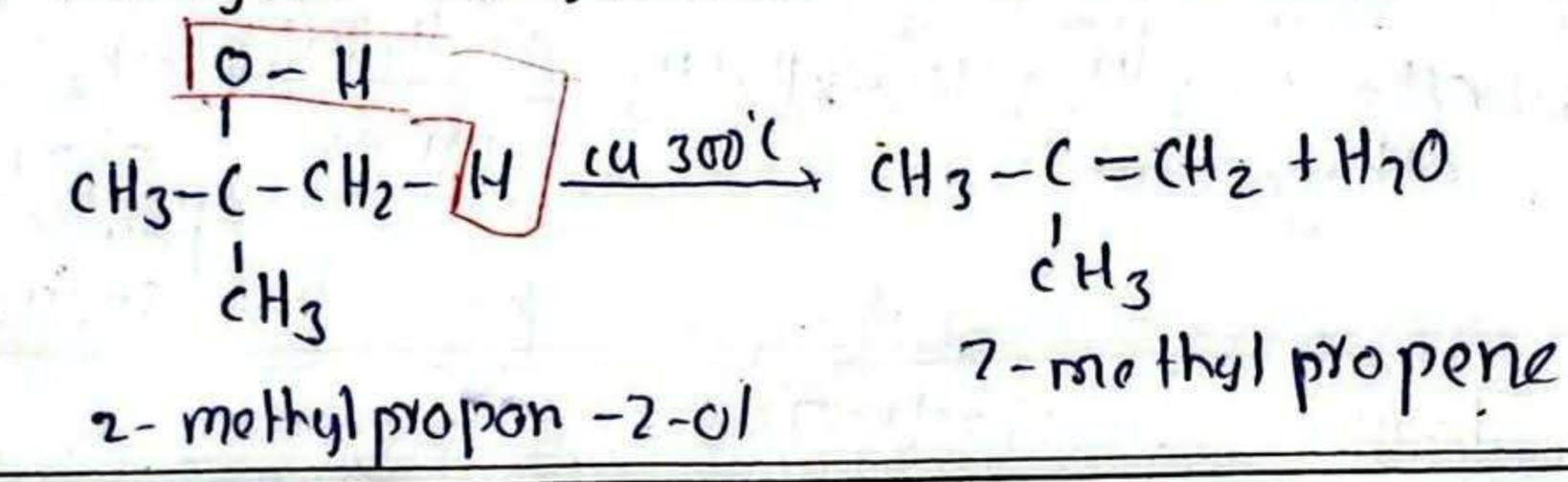
a] catalytic dehydrogenation of 1° alcohols



b] catalytic de-hydrogenation of 2° alcohols



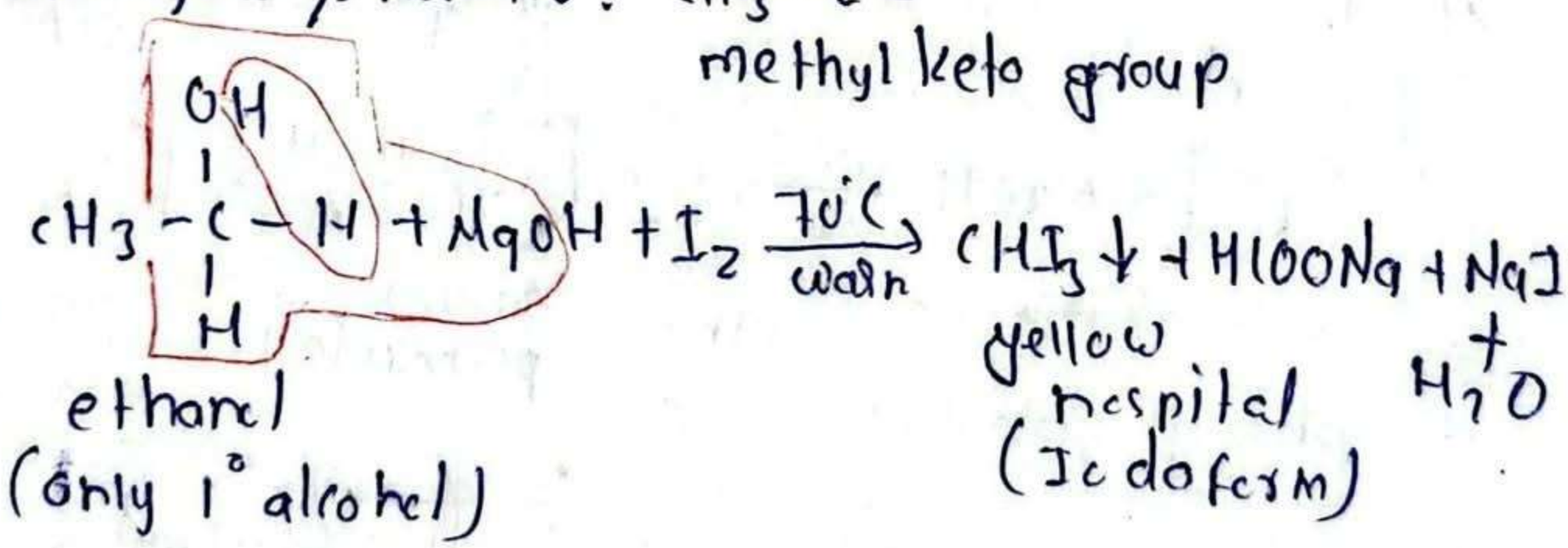
c] Catalytic dehydration of 3° alcohols.

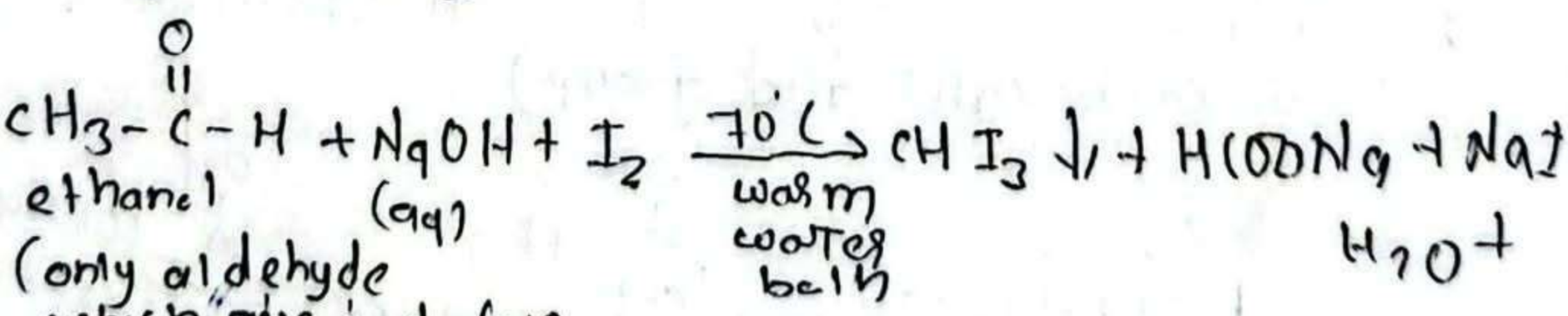


18 Iodoform test

Alcohols: $CH_3-\underset{H}{\overset{OH}{C}}-$

Aldehydes & ketones: $CH_3-\overset{O}{\overset{\|}{C}}-$
methyl keto group



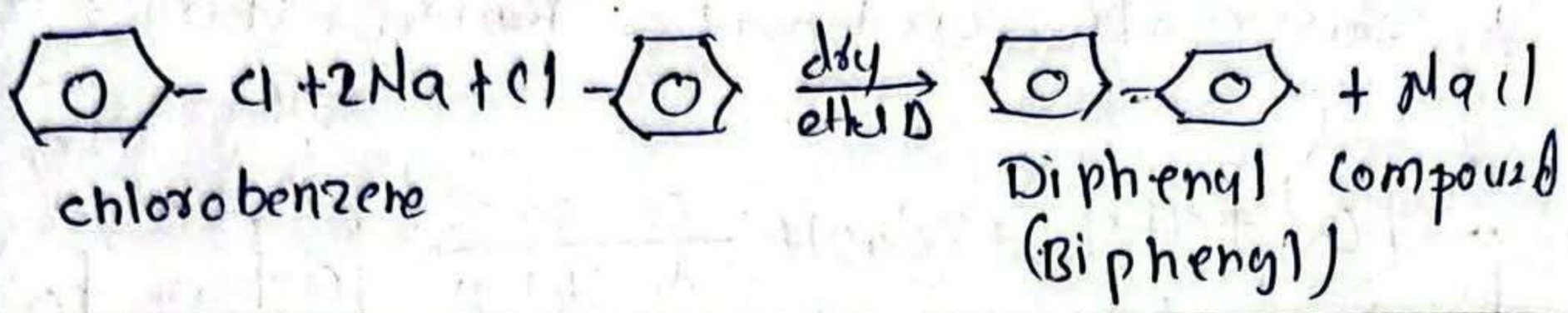


NOTE product formed for ethanol or ethanol in Iodoform test.

Iodofom ১০৭ H কুনা লাই water

$CHI_3 + HCOONa + NaI + H_2O$

13. Fitting reaction :- Helobenzene reacts with sodium metal in the presence of dry ether on heating to give biphenyl compound.



14 Hydroboration :- oxidation of alkenes

$CH_2=CH_2 + (H-BH_2)_3 \xrightarrow[\text{dry ether}]{CCl_2 \text{ or}} CH_2CH_2-BH_2 \xrightarrow[\Delta]{H_2O_2OH} CH_3CH_2OH + H_2BO_3$

Ethene limited — ethyl borane — ethanol

$3\,CH_2=CH_2 + (H-BH_2)_2 \xrightarrow[\text{dry ether}]{CCl_2} (CH_3CH_2)_3B \xrightarrow[\text{"}]{\text{"}} CH_3CH_2OH + H_2BO_3$

excess — triethyl borane — ethanol

15) Reaction of fermentation of molasses (sugar or sucrose) (for detail: see fermentation).

$C_{12}H_{22}O_{11} + H_2O \xrightarrow[H^+]{\text{Invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$

sucrose — glucose — fructose

$C_6H_{12}O_6 \xrightarrow[H^+]{\text{zymase}} 2CH_3CH_2OH + 2CO_2$

glucose or fructose — ethanol.

[16] Dehydration of alcohols :- loss of water from alcohols

a] Dehydration in the presence of excess of conc H_2SO_4

$CH_3-CH_2\boxed{OH-H}\overset{\oplus}{O}\overset{\ominus}{SO_3H} \xrightarrow{100^\circ C} CH_3CH_2-OSO_3H + H_2O$

or H_2SO_4 conc — ethyl hydrogen sulphate

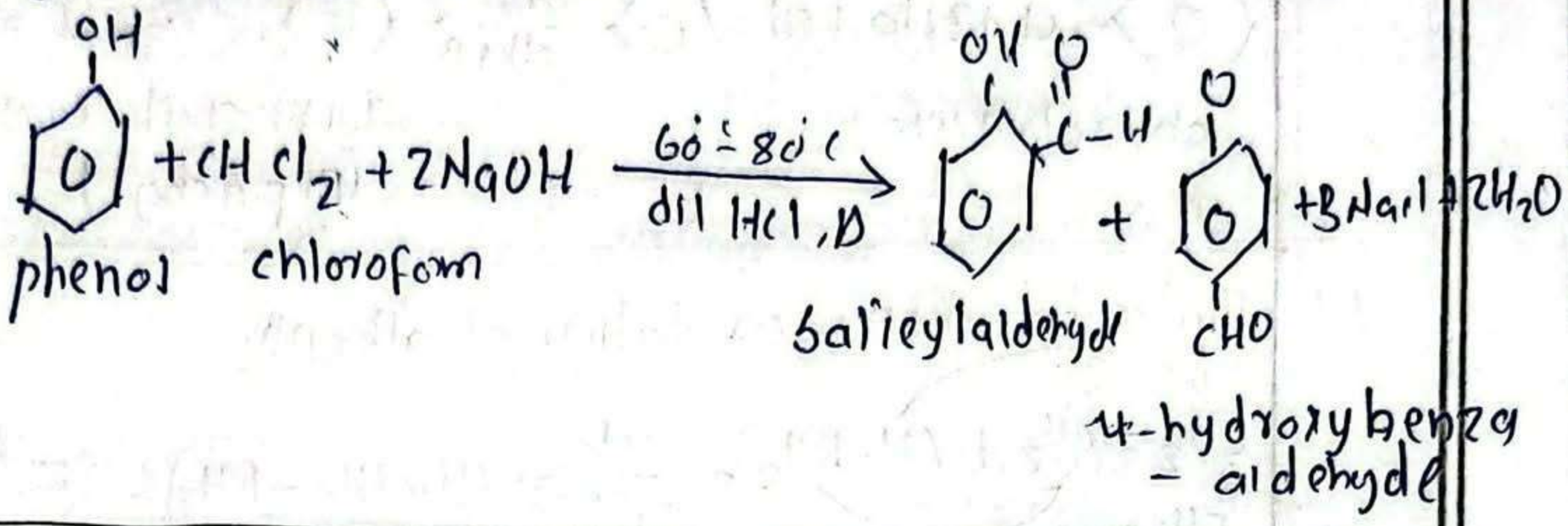
b] Dehydration of alcohols in the presence of Alumina or Silica.

$CH_3CH_2-OH + H-OCH_2CH_3 \xrightarrow[250^\circ C]{SiO_2/Al_2O_3} CH_3CH_2-O-CH_2CH_3 + H_2O$

ethanol — Inter molecular dehydration — ethoxyethane

2082 9] **Reimer - Tiemann's Reaction**

4] When chloroform is heated with phenol in the presence of conc alkali at 60° salicyaldeyde is formed as major product after acidification. This reaction is know as Reimer - Tiemer rexn



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10. **Wurtz reaction** :- When haloalkane is heated with sodium in presence of dry ether, an alkane having double no of carbon atoms than that of haloalk-ane is obtained.

$$R-X + 2Na + X-R \xrightarrow[\text{ether } \Delta]{\text{dry}} R-R + 2NaX$$

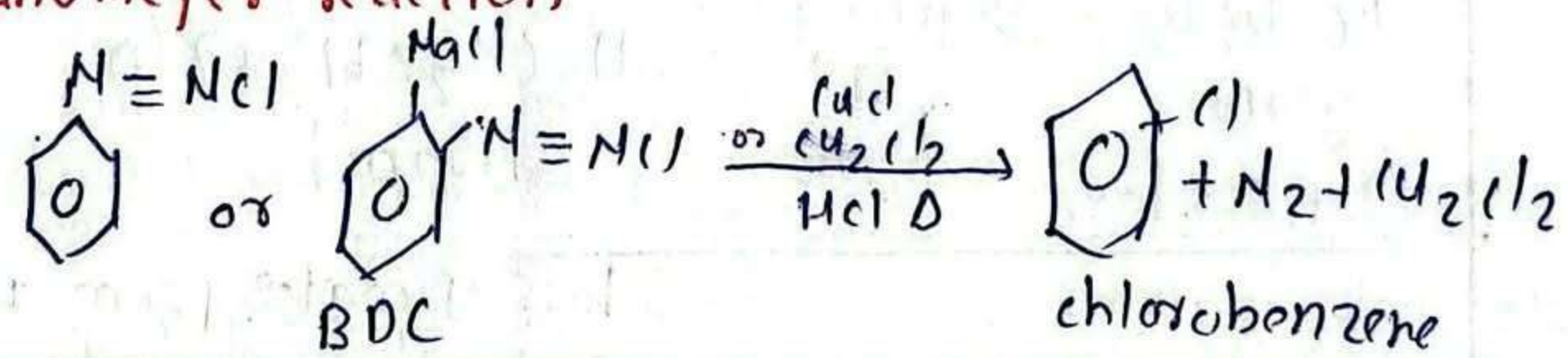
haloalkane → Alkane

$$CH_3-CH(CH_3)-Br + 2Na + Br-CH(CH_3)-CH_3 \xrightarrow[\text{ether}]{\text{dry}} CH_3-CH(CH_3)-CH(CH_3)-CH_3 + 2NaBr$$

2-bromopropane (iso-propyl bromide) → 2,3-dimethy butane

---

11 **Sandmeyer reaction**



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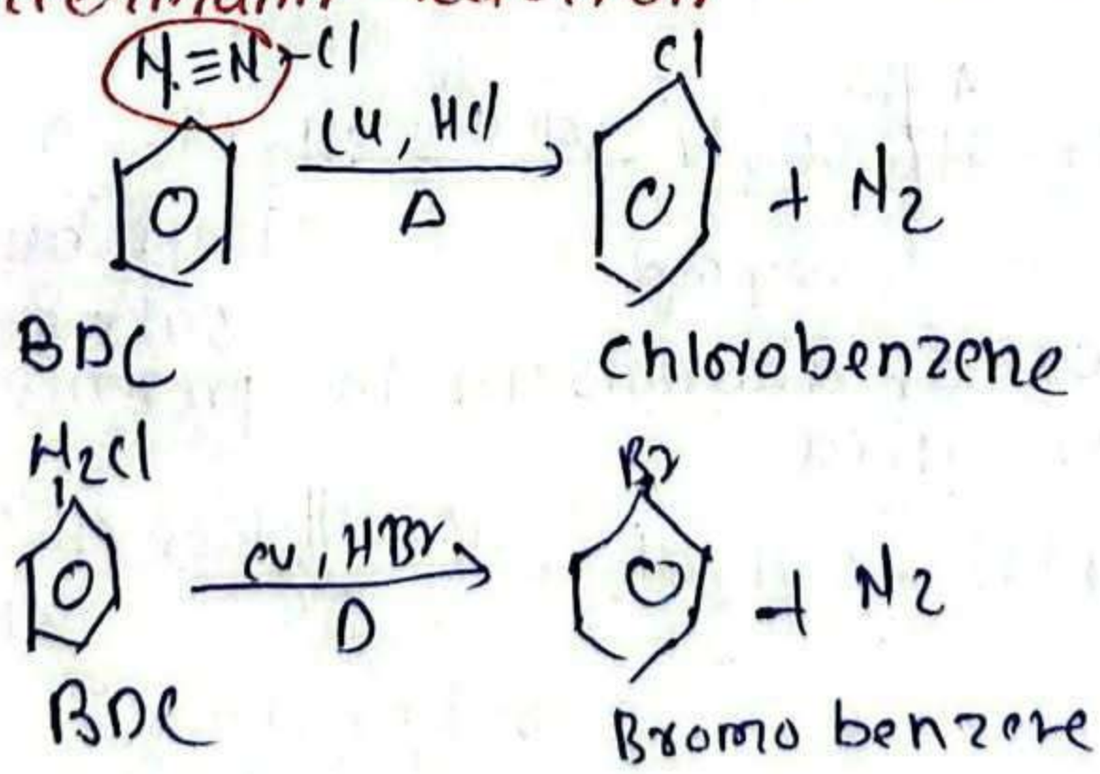
12. **Gattermann reaction**



7] Dehydrohalogenation reaction: α, β-elemination reaction (loss of hydrogen halide)

$$\overset{\beta}{H}-\overset{\beta}{C}H_2-CH_2-Br + alc\ KOH \xrightarrow{\Delta} CH_2=CH_2 + KBr + H_2O$$

Bromoethane (ethyl bromide) → ethane

Saytzeff's rule or Zaitsev rule

$$CH_3-CH_2-\underset{Br}{CH}-CH_3 + alc\ NaOH \xrightarrow{\text{Saytzeff rule}} CH_3-CH=CH-CH_3 + NaBr$$

$$\longrightarrow CH_3-CH_2-CH=CH_2$$

2-bromobutane

$\boxed{3^\circ > 2^\circ > 1^\circ}$ – alkyl halide

NOTE

$$CH_3CH_2Br + KOH\ (aq) \xrightarrow[\text{hydrolysis}]{\Delta} CH_3CH_2OH + KBr$$

Bromethane → ethano

$$CH_3\overset{I}{CH}CH_3 + NaOH\ (aq) \xrightarrow{\Delta} CH_3-\overset{OH}{CH}-CH_3 + NaI$$

2-iodopropane → 2-propanol or propan-2-ol (isopropyl alcohol)

$$CH_3CH_2CH_2-Cl + KOH\ (aq) \xrightarrow{\Delta} CH_3-CH_2-CH_2-OH + KCl$$

1-chloropropane → 1-propanal or propan-2-ol (n-propyl alcohol)

8] Carbylamine test or isocynaide test or isonitrite test (1° amines from 2° 3° amines) or detect chloroform

$$CH_3CH_2NH_2 + CHCl_3 + KOH \xrightarrow{\Delta} CH_3CH_2NC + KCl + H_2O$$

Ethanamine, chloroform, alc

$C_6H_5NH_2$ or $+ CHCl_3 + KOH \xrightarrow[\Delta]{C_2H_5OH} \langle O \rangle NC + H_2O + KCl +$

Aniline (Amino-benzene) (phenylamide) → phenyl isocyanide

NAME REACTIONS

1. Markovnikov rule or Markwonikoff's rule.

$$CH_3-CH=CH_2 + HX \longrightarrow CH_3-\underset{}{\overset{X}{CH}}-CH_3$$

(X = Cl, Br, I) 2-halopropane

2. Anti-Markovnikov's rule or peroxide effect or kharash effect

$$CH_3-\overset{CH_3}{C}=CH-CH_3 + HBr \xrightarrow[R-O-O-R']{\text{organic peroxide}} CH_3-\overset{CH_3}{CH}-\overset{Br}{CH}-CH_3$$

2-methyl but-2ene → 2-Bromo-3 methyl butane

3. Darzen's reaction

$$ROH + SOCl_2 \xrightarrow{\text{Reflux}} R-Cl + SO_2\uparrow + HCl\uparrow$$

Alcohol → chloroalkane

4 Hoffmann's ammonolysis of alkyl halides

$$CH_3CH_2Br + H-NH_2 \xrightarrow[100^\circ C]{\Delta} CH_3CH_2-NH_2 + HBr$$

Bromethane; (alc) or NH_3; Ethanomine (1° amine)

Fix 5 Williamson's etherification or williamson ether synthesis

↳ When alkyl halide is heated with sodium alkoxide, or sodium phenoxide ether is formed. This reaction is known as williamson's etherification (S^2 mechanism)

$$CH_3-ONa + Cl-CH_3 \xrightarrow{\Delta} CH_3-O-CH_3 + NaCl$$

sodium methaxide; methy-chloride; methoxymethane

Aliphatic symmetrical or simple ethanes

6] Mendius reduction

$$-CN + 4[H] \xrightarrow[\Delta]{Na/C_2H_5OH}$$

$$-CH_2NH_2$$

Benzenenil-rile (Benzonitrile) → ethanamine ~~(Ethyl amide)~~

NOTE) Dry ether should be used during the preparation of Grignard's reagents and during the reactions of Grignarides reagent. If moist ether is used moisture react with

Grignard's reagent forming corresponding alkanes.

$$\overset{\delta+}{CH_3}\overset{\delta-}{Br} + Mg \xrightarrow{ether} CH_3MgBr$$

methyl Bromide — methyl magnesium bromide

$$\overset{\delta-}{CH_3}\boxed{\overset{\delta+}{Mg}Br + HO}-H \xrightarrow{\Delta} CH_4 + Mg(OH)Br$$

methane

Q. Reaction of Grignard's reagents with (O_2 or oxygen (air))

During the preparation and reaction of Grignard's reagent dry ether is used as solvent and as inert medium to avoid the reaction of Grignard's reagent in exposed in air.

$$CH_3MgBr + C(=O)-O \xrightarrow[ether]{dry} CH_3-C(=O)-O\boxed{MgBr} \xrightarrow[\Delta]{H-O-H} CH_3-C(=O)-OH + Mg(OH)Br$$

methyl mag. bromide — ethanoic acid

100% 5 mark

Grignard reagents

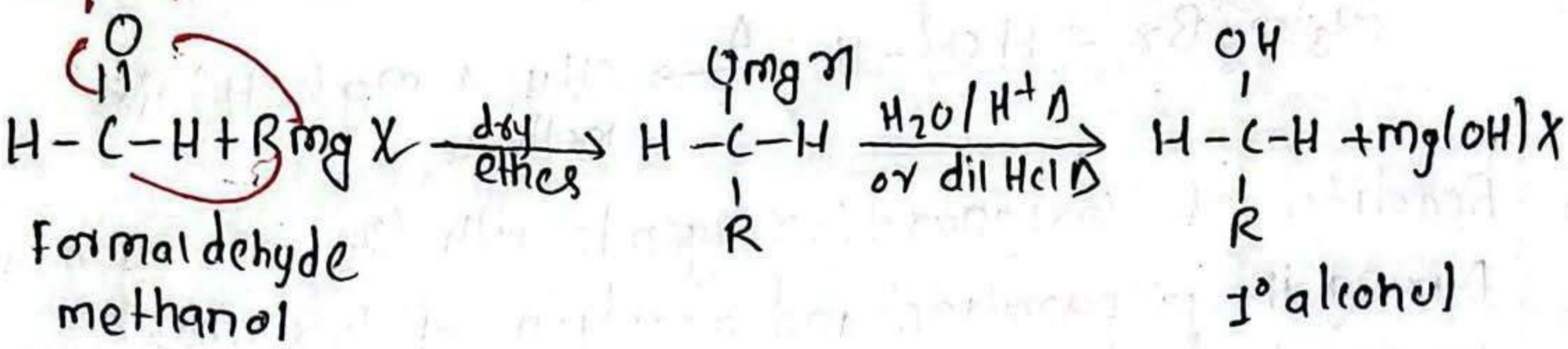
$$R-X + Mg \xrightarrow{\text{dry ether}} R-MgX$$

Alkyl halide → Alkyl magnesium halide

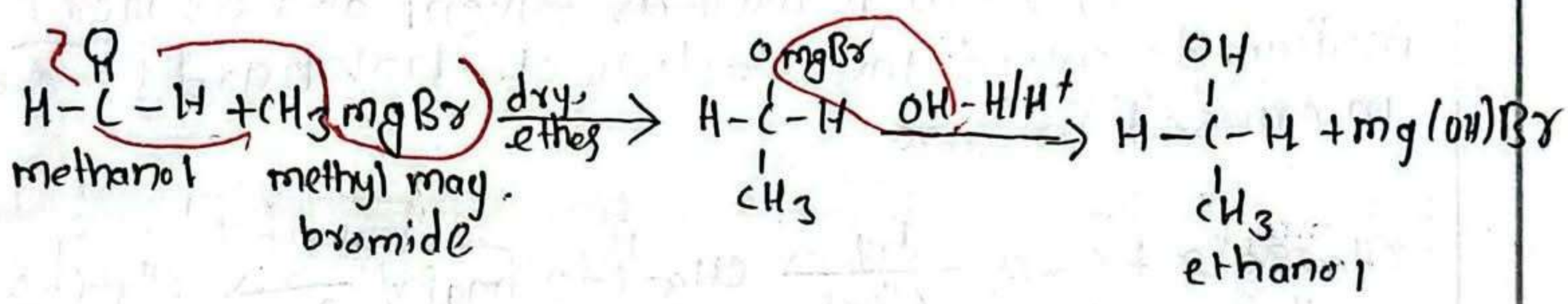
$$CH_3CH_2-Br + Mg \xrightarrow{\text{dry ether}} CH_3CH_2MgBr$$

Ethyl bromide → ethyl mag. bromide

a] preparation of 1° alcohols

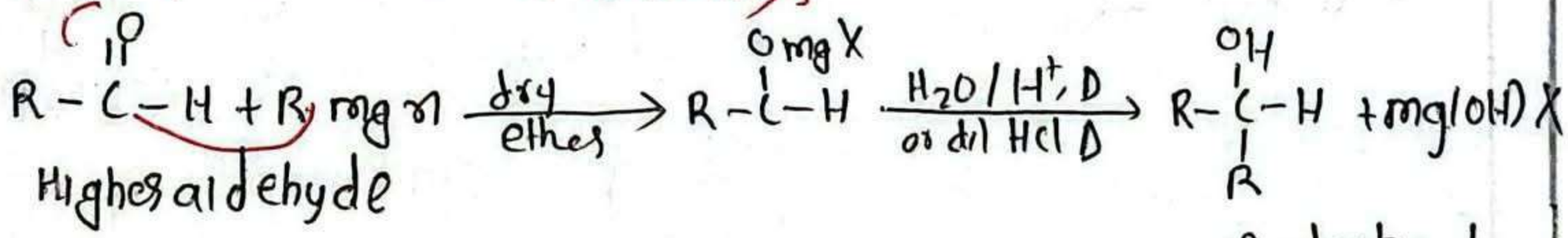


Formaldehyde → methanol; 1° alcohol

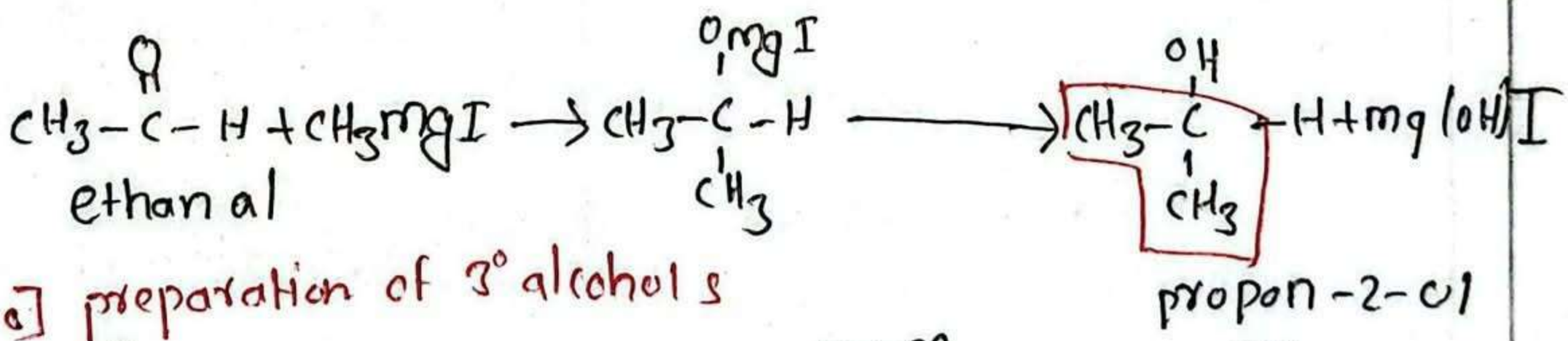


methanol; methyl mag. bromide; ethanol

b] preparation of 2° alcohols

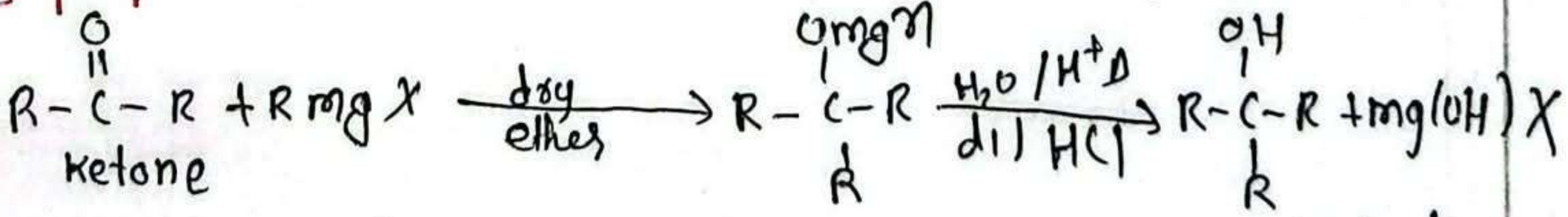


Higher aldehyde; 2° alcohol

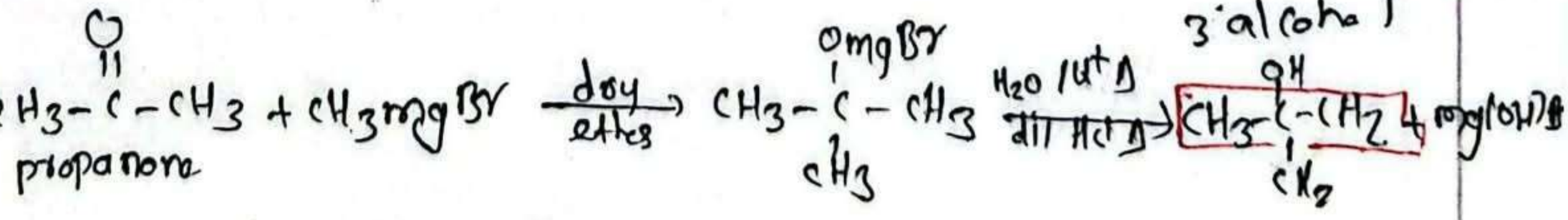


ethanal; propan-2-ol

c] preparation of 3° alcohols



Ketone; 3° alcohol



propanone

2] Methanol to ethanol and vice versa

→ Soln

$CH_3OH \xrightarrow{PCl_3} CH_3Cl \xrightarrow{alc\ KCN} CH_3CN$

methanol — chloromethane — ethanenitrile

$\downarrow$ 4[H] / H_2

$CH_3CH_2OH \xleftarrow{HNO_2} CH_3CH_2NH_2$

ethanol — Ethanamide

again,

$CH_3CH_2OH \xrightarrow[[O]]{KMnO_4/H^+} CH_3COOH \xrightarrow{NH_3} CH_3COONH_4$

ethanol — ethanoic acid — Ammonium ethanoate

$\downarrow -H_2O$

$CH_3OH \xleftarrow{HNO_2} CH_3NH_2 \xleftarrow{Br_2/KOH} CH_3CONH_2$

methanol — ~~ethanamide~~ methamide — ethanamide

③ 1-chloro propane into 2-chloro propane

$CH_3CH_2CH_2-Cl \xrightarrow{(KOH)} CH_3-CH=CH_2 \xrightarrow{HCl} CH_3-\underset{Cl}{CH}-CH_3$

1-chloropropane — propen gas — 2-chloropropane

100% Q How is trichloromethane prepared in lab from acetone and ethanol?

→ lab preparation of trichloromethane.

Theory (principle)

chloroform is prepared in lab by heating bleaching powder past with ethanol or acetone. Bleaching powers acts as chlorinating oxidizing & hydrolyzing agent.

$$CaOCl_2 + H_2O \longrightarrow Cl_2 + Ca(OH)_2$$

① From ethyl alcohol (Ethanol)

a] The oxidications of Ethyl gives Acetaldehyde

$$CH_3CH_2OH + Cl_2 \xrightarrow{\Delta} CH_3-\overset{O}{\overset{\|}{C}}-H + 2HCl$$

ethanol (Acetaldehyde)

b] The chlorazation of Acetaldehyded give trichlorid Acetaldehyde

$$CH_3-\overset{O}{\overset{\|}{C}}-H + 3Cl_2 \longrightarrow CCl_3-\overset{O}{\overset{\|}{C}}-H + 3HCl$$

chloral

c] The hydrolysises of trichloro acetaldehyde gives chloro form

$$CCl_3-\overset{O}{\overset{\|}{C}}-H + Ca(OH)_2 \xrightarrow{\Delta} CHCl_3 + (HOO)_2Ca$$

chloroform Cal. form

② for Acetone (propone)

a] The chlorozations of Acetone give trichloro acetone

$$CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + 3Cl_2 \xrightarrow{\Delta} CCl_3-\overset{O}{\overset{\|}{C}}-CH_3 + 3HCl$$

propanone trichloro-propane

b] The hydrolyles trichloro acetone give chloroform

$$2CCl_3-\overset{O}{\overset{\|}{C}}-CH_3 + Ca(OH)_2 \xrightarrow{\Delta} 2CHCl_3 + (CH_3COO)_2Ca$$

Cal acetate

2] Convert Chloroform into

a] Ethyne

$$\underset{\text{chloroform}}{2CHCl_3} + 6Ag \xrightarrow{-(\Delta)} \underset{\text{Ethyne}}{HC\equiv CH} + \underset{\text{silver chloride}}{6AgCl}$$

b] chloropicrin

$$\underset{\text{chloroform}}{CHCl_3} + \underset{\text{Nitric acid}}{HNO_3\ (\text{conc})} \xrightarrow{-(\Delta)} CCl_3NO_2 + \underset{\text{water}}{H_2O}$$

c] chloretone

$$\underset{\text{chloroform}}{CHCl_3} + \underset{\text{Acetone}}{C_3H_6O} \xrightarrow{KOH/H_2O} \underset{\text{chloretone}}{C_4H_7Cl_3O}$$

d] formic acid

$$\underset{\text{chloroform}}{CHCl_3} + 4KOH \xrightarrow{\Delta} \underset{\text{formic acid (HCOOH)}}{HCOOK} + 3KCl + 2H_2O$$

Q. Convert

a] 1-propanol to 2-propanal and vice versa

$$\underset{\text{1-propanol}}{CH_3CH_2CH_2OH} \xrightarrow[170^\circ C]{\text{conc } H_2SO_4} \underset{\text{propene}}{CH_3-CH=CH_2} + HBr \longrightarrow \underset{\text{2-chloro propane}}{CH_3-\overset{Cl}{\overset{|}{CH}}-CH_3}$$

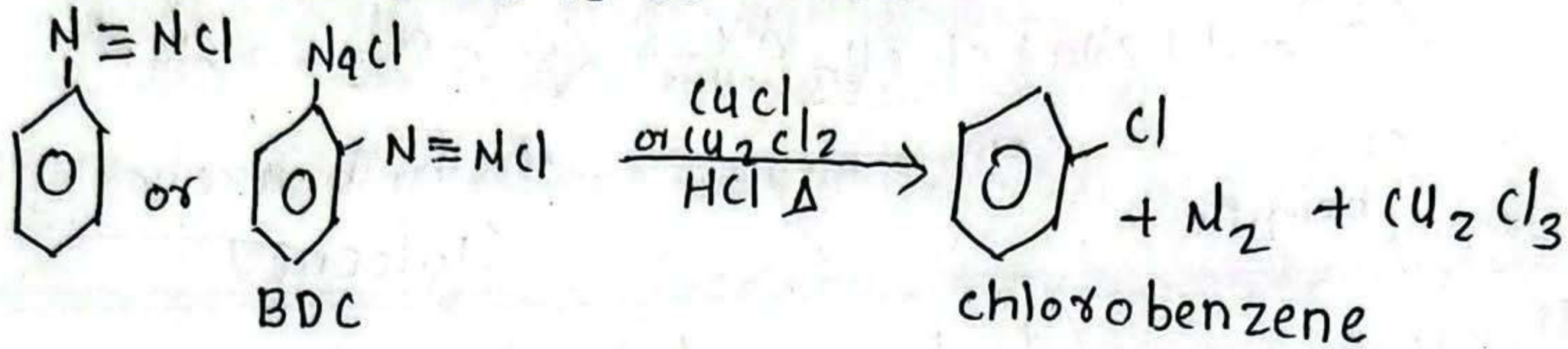
$$\xrightarrow{aqNaOH} \underset{\text{2-propanal}}{CH_3-\underset{OH}{\underset{|}{CH}}-CH_3}$$

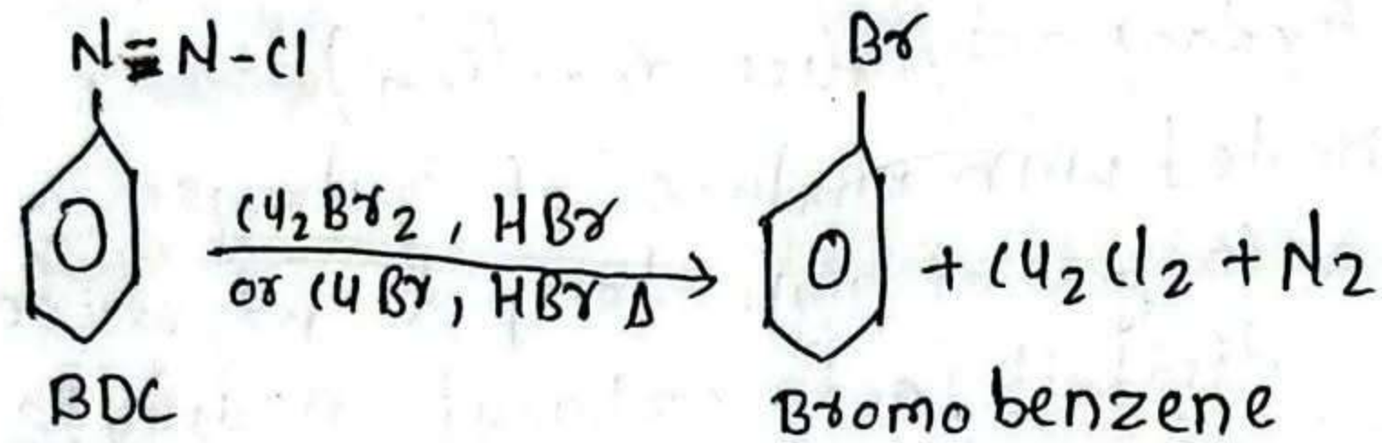
$$\underset{\text{2-propane}}{CH_3-\overset{OH}{\overset{|}{CH}}-CH_3} \xrightarrow[170^\circ]{\text{conc } H_2SO_4} \underset{\text{propene}}{CH_3-CH=CH_2} + HBr \xrightarrow{H_2O_2} \underset{\text{1-bromo propane}}{CH_3-CH_2-CH_2Br}$$

$$\xrightarrow[KOH]{NaOH} \underset{\text{1-propanol}}{CH_3CH_2CH_2OH}$$

Chemistry Important Questions (NEB)

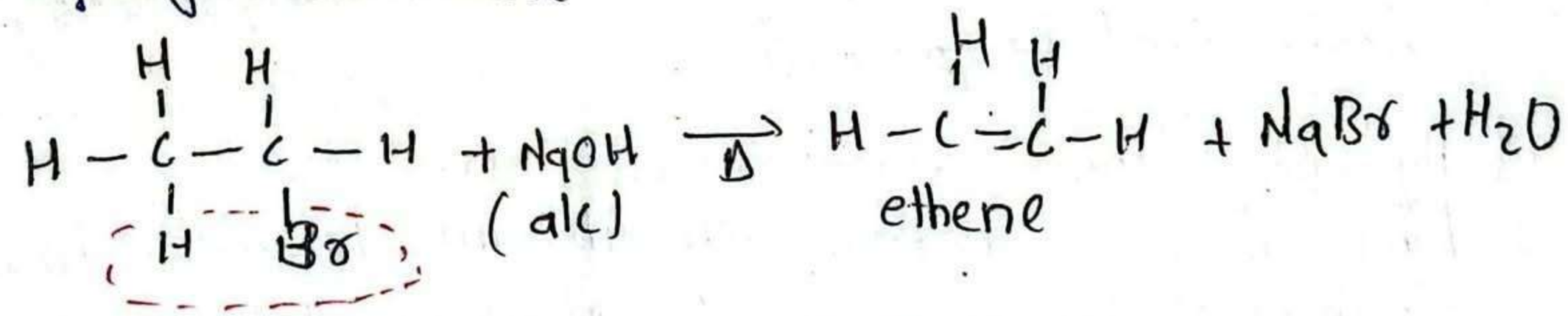
(1) Sandmeyer's reaction :- Benzendiazonium chloride when warmed with cuprous chloride in the presence of HCl or cuprous bromide & HBr chlorobenzene or bromo-benzene is obtained.

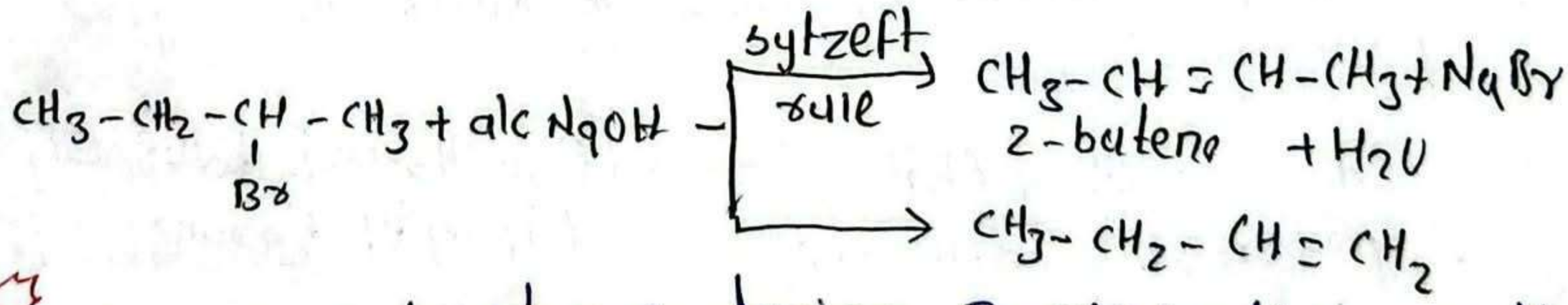




2) De-hydrohalogenation reaction

→ When alkyl haiide is heated with alkolic soln. of NaOH or KOH, Alkene is obtained by dy de-hydrohalogenation reaction, this reaction is also know as β-elimination, this reacion is also know as β-elimination reaction as β-hydrogen are remove





NOTE Accoding to Sytzeft rue, during β-elimination reaction highly branch or highly subsitutes alkene is favourable.